

Борис Копее Очерков

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$$z = (x, y) = x + iy$$

$$x, y \in \mathbb{R}, \quad x = \operatorname{Re} z, \quad y = \operatorname{Im} z$$

x	1	i
1	1	i
i	i	-1



$$z_1 = x_1 + y_1 i, \quad z_2 = x_2 + y_2 i$$

$$z_1 z_2 = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + x_2 y_1)$$

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$\mathbb{C}$ - это поле

$$z \neq 0 \cdot 0 + i \cdot 0, \quad \text{но} \quad z^{-1} = \frac{\bar{z}}{|z|^2}$$

$$\frac{z \bar{z}}{|z|^2} = 1$$

$$\bar{z} = x - yi$$

$$|z| = \sqrt{x^2 + y^2}$$

$$|z|^2 = z \bar{z} = x^2 + y^2$$

$$z_1 z_2 = 1$$

$$\begin{cases} x_1 x_2 - y_1 y_2 = 1 \\ x_1 y_2 + x_2 y_1 = 0 \end{cases}$$

$$\begin{vmatrix} x_1 & -y_1 \\ y_1 & x_1 \end{vmatrix} = x_1^2 + y_1^2 \neq 0 \quad z \neq 0$$

$$z \neq 0$$

$$z = r(\cos \varphi + i \sin \varphi)$$

мпол. форма гамили

$$r > 0$$

$$r = |z| = \sqrt{z \bar{z}}, \quad \varphi = \arg z$$

$$\operatorname{tg} \varphi = \frac{y}{x}$$

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$$

$$\operatorname{Arg} z = \{\arg z + 2\pi n, n \in \mathbb{Z}\}$$

$$\arg_{\operatorname{rn}} z \in (-\pi, \pi] \quad \text{наиболь. глос. арг.}$$

$$z_1 = r_1(\cos \varphi_1 + i \sin \varphi_1)$$

$$z_2 = r_2(\cos \varphi_2 + i \sin \varphi_2)$$

$$z_1 z_2 = r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2))$$

$$|z_1 z_2| = |z_1| |z_2|$$

$$\operatorname{Arg}(z_1 z_2) = \operatorname{Arg} z_1 + \operatorname{Arg} z_2$$

$$e^{i\varphi} = \cos \varphi + i \sin \varphi \quad \Rightarrow \quad z = r e^{i\varphi}$$

$$z^n = r^n e^{in\varphi}, \quad n \in \mathbb{Z} \quad \text{формула Муавра}$$

$$z^{-1} = \frac{\bar{z}}{|z|^2} = \frac{r(\cos\varphi + i\sin\varphi)}{r^2} = \frac{1}{r}(\cos\varphi - i\sin\varphi) = r^{-1}e^{-i\varphi}$$

$$z = x + iy \iff \begin{pmatrix} x & -y \\ y & x \end{pmatrix} \in M_{2 \times 2}(\mathbb{R})$$

$$z_1 = x_1 + iy_1$$

$$z_2 = x_2 + iy_2$$

$$\begin{pmatrix} x_1 & -y_1 \\ y_1 & x_1 \end{pmatrix} \begin{pmatrix} x_2 & -y_2 \\ y_2 & x_2 \end{pmatrix} = \begin{pmatrix} x_1 x_2 - y_1 y_2 & -x_1 y_2 - x_2 y_1 \\ y_1 x_2 + x_1 y_2 & x_1 x_2 - y_1 y_2 \end{pmatrix}$$

$$\begin{pmatrix} x & -y \\ y & x \end{pmatrix} = \begin{pmatrix} r\cos\varphi & -r\sin\varphi \\ r\sin\varphi & r\cos\varphi \end{pmatrix} = r \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix} \begin{matrix} z_1 z_2 \\ \end{matrix}$$

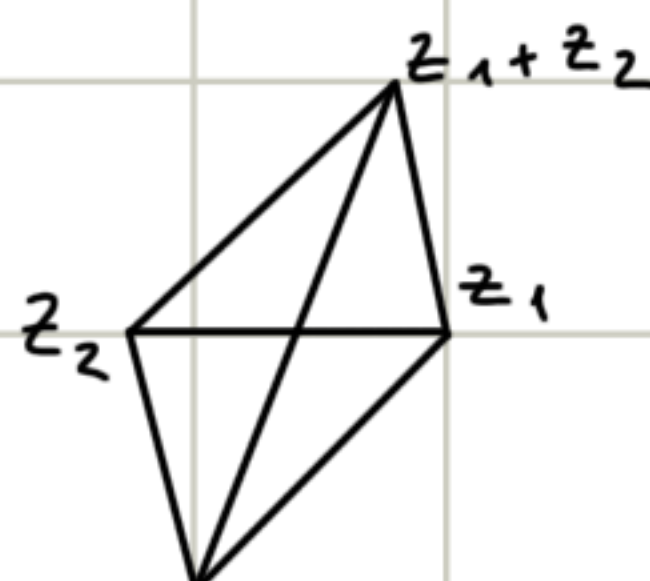
$$z^{-1} \iff r^{-1} \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix}$$

метрика

$$\rho(z_1, z_2) = |z_1 - z_2|$$

$$|z_1| + |z_2| \geq |z_1 + z_2|$$

$$|z_1 - z_2| \geq ||z_1| - |z_2||$$



$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) + (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) = z_1 \bar{z}_1 + z_2 \bar{z}_2 + z_1 \bar{z}_2 + z_2 \bar{z}_1 + z_1 \bar{z}_1 + z_2 \bar{z}_2 - z_2 \bar{z}_1 - z_1 \bar{z}_2 = 2|z_1|^2 + 2|z_2|^2$$

$$+ z_1 \bar{z}_1 + z_2 \bar{z}_2 - z_2 \bar{z}_1 - z_1 \bar{z}_2 = 2|z_1|^2 + 2|z_2|^2$$

$$\bar{\mathbb{C}} = \mathbb{C} \cup \{\infty\} \quad \text{расширенная комплексная пл-ть}$$

$$z_0 \in \mathbb{C} \quad U_\varepsilon(z_0) = \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}$$

$$\overset{\circ}{U}_\varepsilon(z_0) = U_\varepsilon(z_0) \setminus \{z_0\}$$

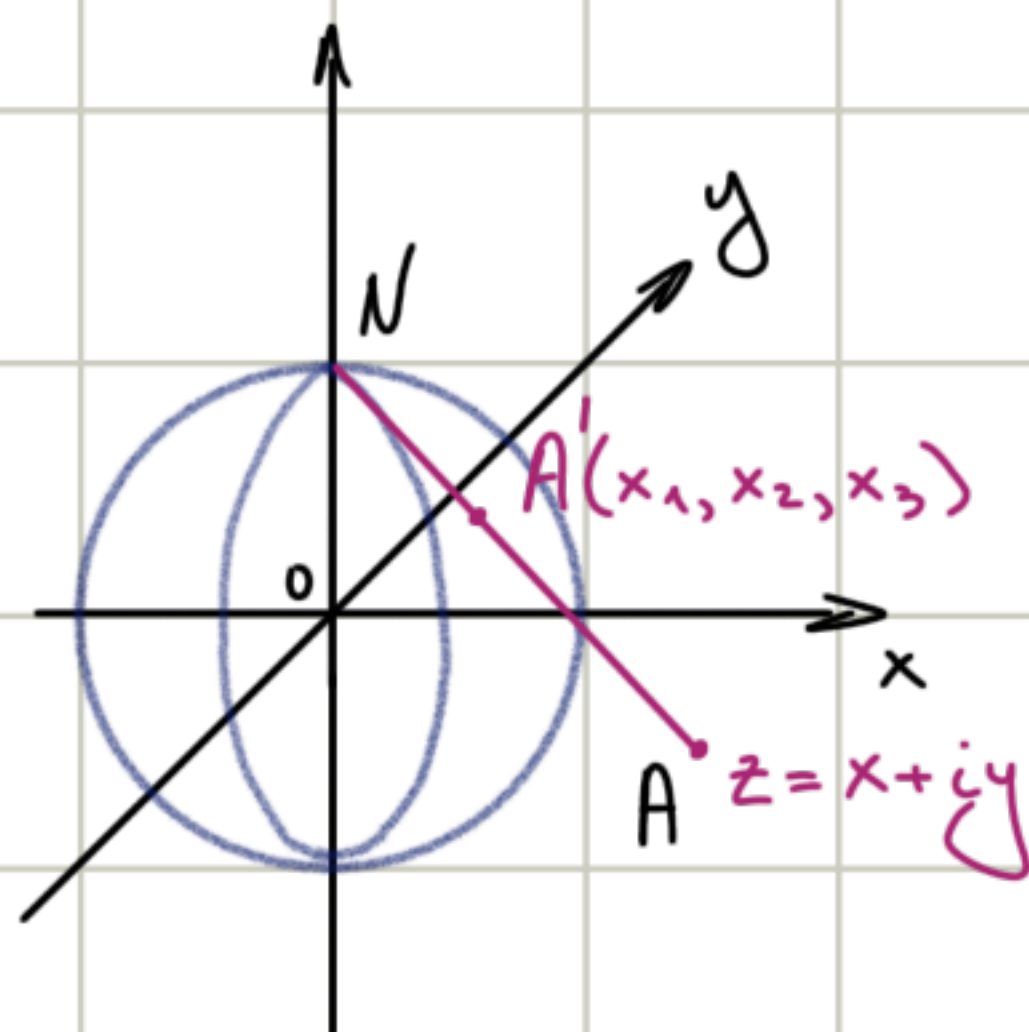
$$\overset{\circ}{U}_\varepsilon(\infty) = \{z \in \mathbb{C} : |z| > \varepsilon\}$$

$$z_n \in \mathbb{C}$$

$$\lim_{n \rightarrow \infty} z_n = \infty, \text{ если } \forall \varepsilon > 0 \exists N \in \mathbb{N} \quad \forall n > N \hookrightarrow z_n \in \overset{\circ}{U}_\varepsilon(\infty)$$

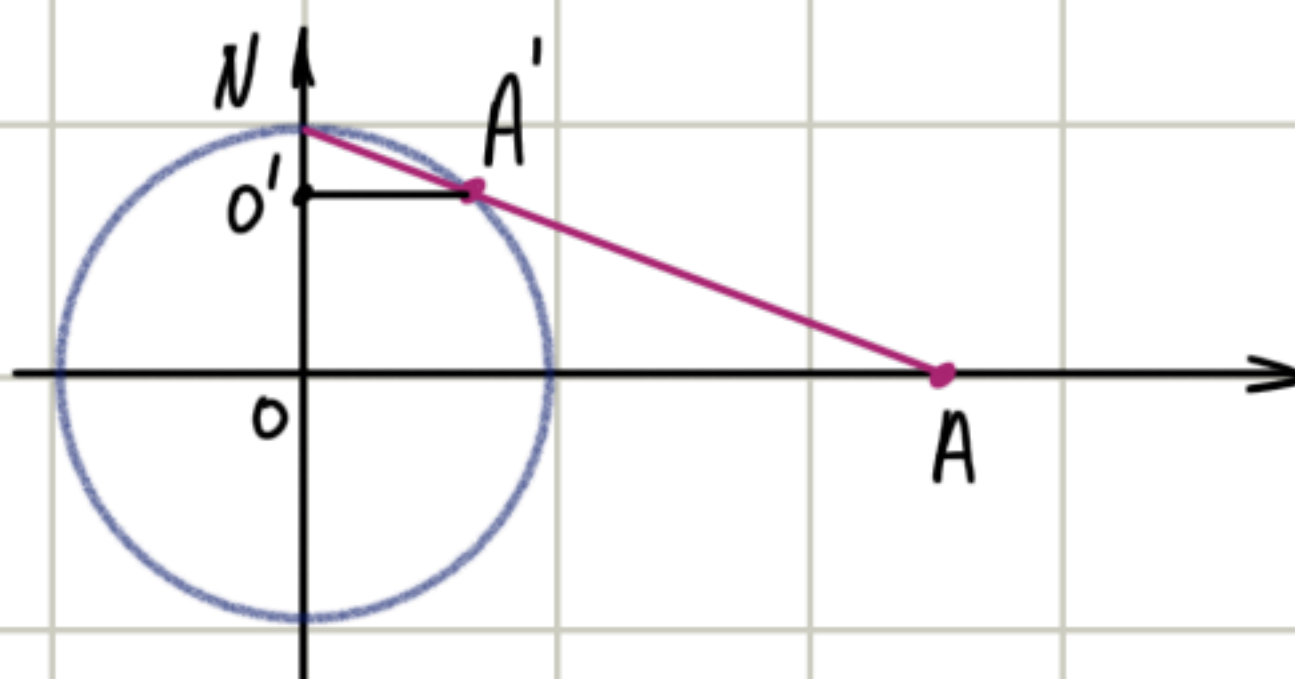
$$\Leftrightarrow \lim_{n \rightarrow \infty} |z_n| = \infty$$

$$\text{Если } z_0 \in \mathbb{C} \quad \lim_{n \rightarrow \infty} z_n = z_0 \Leftrightarrow \lim_{n \rightarrow \infty} |z_n - z_0| = 0$$



Стереосферическая проекция

$N = (0, 0, 1)$ - северный полюс



$$\frac{|NO'|}{|NO|} = \frac{|O'A'|}{|OA|}$$

$$\Delta NO'A' \sim \Delta NOA$$

$$\frac{1-x_3}{1} = \frac{\sqrt{x_1^2 + x_2^2}}{|z|} = \frac{\sqrt{1-x_3^2}}{|z|}$$

$$|z| = \frac{\sqrt{1-x_3^2}}{1-x_3} = \sqrt{\frac{(1+x_3)(1-x_3)}{(1-x_3)(1-x_3)}} = \sqrt{\frac{1+x_3}{1-x_3}}$$

$$|z|^2 = \frac{1+x_3}{1-x_3}$$

$$|z|^2(1-x_3) = (1+x_3)$$

$$x_3(|z|^2 + 1) = |z|^2 - 1$$

$$x_3 = \frac{|z|^2 - 1}{|z|^2 + 1}$$

$$\frac{x_1}{x_2} = x$$

$$x_1^2 + x_2^2 = 1 - x_3^2 = 1 - \frac{(|z|^2 - 1)^2}{(|z|^2 + 1)^2} = \frac{(|z|^2 + 1)^2 - (|z|^2 - 1)^2}{(|z|^2 + 1)^2} = \frac{4|z|^2}{(|z|^2 + 1)^2}$$

$$x_1 + x_2 i = \frac{z|z|(x+yi)}{(|z|^2 + 1)|z|} = \frac{z}{(|z|^2 + 1)}$$

$$\begin{cases} x_1 = \frac{z + \bar{z}}{(1+|z|^2)} = \operatorname{Re} \frac{z}{(|z|^2 + 1)} \\ x_2 = \frac{z - \bar{z}}{(1+|z|^2)} = \operatorname{Im} \frac{z}{(|z|^2 + 1)} \\ x_3 = \frac{|z|^2 - 1}{|z|^2 + 1} \end{cases}$$

Проекция прямой - это окружности на сфере, проходящие через северный полюс

Конформное отображение - отображение, сохраняющее углы между кривыми

Стереогр. проекция - конформн. отображ.

Область в $\mathbb{C}$ - непустое открытое связное мн-во

A - связное мн-во в $\mathbb{C}$, если $\exists A_1, A_2$ - открытых точек, что

$$A = (A \cap A_1) \cup (A \cap A_2), \quad \text{причем } A \cap A_1 \neq \emptyset, \quad A \cap A_2 \neq \emptyset$$

$C = A \cup B$ - дизъюнктное объединение, если $C = A \cup B$ и $A \cap B = \emptyset$

$\exists f: \mathcal{D} \rightarrow \bar{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. $\mathcal{D}$ - область

$\exists z_0 \in \mathcal{D}$

$\lim_{z \rightarrow z_0} f(z) = A \in \bar{\mathbb{C}}$, если $\forall \varepsilon > 0 \quad \exists \delta > 0$

$$\forall z \in \dot{U}_\delta(z_0) \rightarrow f(z) \in U_\varepsilon(A)$$

$$f(z) = u(x, y) + i v(x, y), \quad \text{где } u = \operatorname{Re} f$$

$$v = \operatorname{Im} f$$

$$f(z) \leftrightarrow \begin{pmatrix} u \\ v \end{pmatrix} \quad \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

Def.

$\exists f: U_{\delta}(z_0) \rightarrow \mathbb{C}$. Ф-я f наз. $\mathbb{R}$ -гифф-мой в точке z_0 , если $\begin{pmatrix} u(x,y) \\ v(x,y) \end{pmatrix}$ - гифф-ма в м. (x_0, y_0) , где $z_0 = x_0 + iy_0$

$$\begin{pmatrix} u(x,y) - u(x_0, y_0) \\ v(x,y) - v(x_0, y_0) \end{pmatrix} = D_f(x_0, y_0) \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} + o(\sqrt{(x-x_0)^2 + (y-y_0)^2}) = D_f(x_0, y_0) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} + o(\sqrt{\Delta x^2 + \Delta y^2})$$

$\begin{matrix} x \rightarrow x_0 \\ y \rightarrow y_0 \end{matrix}$ $\begin{matrix} \Delta x \rightarrow 0 \\ \Delta y \rightarrow 0 \end{matrix}$

$$D_f(x_0, y_0) = \begin{pmatrix} \frac{\partial u}{\partial x}(x_0, y_0) & \frac{\partial u}{\partial y}(x_0, y_0) \\ \frac{\partial v}{\partial x}(x_0, y_0) & \frac{\partial v}{\partial y}(x_0, y_0) \end{pmatrix} \quad \text{матр. Якоби}$$

Def.

$\exists f: U_{\delta}(z_0) \rightarrow \mathbb{C}$. Ф-я f наз. $\mathbb{C}$ -гифф-мой (гифф. в компл-м смысле),

если $\exists A \in \mathbb{C}$: $f(z + \Delta z) - f(z) = A \cdot \Delta z + o(\Delta z)$, $\Delta z \rightarrow 0$

$\alpha(\Delta z) = o(\Delta z)$, $\Delta z \rightarrow 0$, если $\lim_{\Delta z \rightarrow 0} \frac{\alpha(\Delta z)}{\Delta z} = 0 \Leftrightarrow \lim_{\Delta z \rightarrow 0} \frac{|\alpha(\Delta z)|}{|\Delta z|} = 0$

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{комплекс. проиув. ф-и f в м. z_0 }$$

Предложение

f - $\mathbb{C}$ -гифф. в м. $z_0 \Leftrightarrow \exists f'(z_0)$

$\Delta \Rightarrow \exists f$ - $\mathbb{C}$ -гифф. в м. z_0 , тогда $f(z) - f(z_0) = A(z - z_0) + o(z - z_0)$ $z \rightarrow z_0$

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{z \rightarrow z_0} \frac{A(z - z_0)}{z - z_0} + \lim_{z \rightarrow z_0} \frac{o(z - z_0)}{z - z_0} = A = f'(z_0)$$

$$\Leftarrow \exists f'(z_0) = A = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \Leftrightarrow A = \frac{f(z) - f(z_0)}{z - z_0} + \alpha(z), \text{ где}$$

$$\lim_{z \rightarrow z_0} \alpha(z) = 0 \Rightarrow f(z) - f(z_0) = A(z - z_0) + \underbrace{\alpha(z)(z - z_0)}_{o(z - z_0)} \quad z \rightarrow z_0$$

th

$\exists f: U_{\delta}(z_0) \rightarrow \mathbb{C}$

$f = u + vi$, $z_0 = x_0 + iy_0$, ф-я f - $\mathbb{C}$ -гифф. в м. $z_0 \Leftrightarrow$ ф-я f $\mathbb{R}$ -гифф. в м. z_0 и

вып. усл. Коши-Римана:

$$\begin{cases} \frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0) = \alpha \\ \frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0) = \beta \end{cases}$$

матр. Якоби $D_f = \begin{pmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$

$\Delta \Rightarrow \exists f\text{-}\mathbb{C}\text{-guarop. b m. } z_0$

$$f(z+\Delta z) - f(z) = f'(z_0)\Delta z + \bar{o}(\Delta z), \quad \Delta z \rightarrow 0 \quad (*)$$

$$f'(z_0) = \alpha + \beta i, \quad \alpha, \beta \in \mathbb{R}, \quad \text{wge } g(\Delta z) = \bar{o}(\Delta z) \quad \Delta \rightarrow 0$$

$$\Delta z = \Delta x + \Delta y i$$

$$(*) \Leftrightarrow \begin{aligned} 1) & u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) = \operatorname{Re}(f'(z_0)\Delta z + \bar{o}(\Delta z)) = \\ & = \operatorname{Re}((\alpha + \beta i)(\Delta x + \Delta y i) + g_1(\Delta x, \Delta y)) = \alpha \Delta x - \beta \Delta y + g_1(\Delta x, \Delta y) \\ 2) & v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0) = \operatorname{Im}(f'(z_0)\Delta z + \bar{o}(\Delta z)) = \operatorname{Im}((\alpha + \beta i)(\Delta x + \Delta y i) + g_2(\Delta x, \Delta y)) = \beta \Delta x + \alpha \Delta y + g_2(\Delta x, \Delta y) \end{aligned}$$

$$\begin{aligned} \text{wge } g_1(\Delta z) &= \operatorname{Re} g(\Delta z) \\ g_2(\Delta z) &= \operatorname{Im} g(\Delta z) \end{aligned}$$

$$v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0) = \beta \Delta x + \alpha \Delta y + g_2(\Delta x, \Delta y)$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{g_1(\Delta x, \Delta y)}{\sqrt{\Delta x^2 + \Delta y^2}} = 0, \quad |g_1(\Delta x, \Delta y)| \leq |g(\Delta x, \Delta y)|$$

$$|g_2(\Delta x, \Delta y)| \leq |g(\Delta x, \Delta y)|$$

$$\text{more } \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{|g_1(\Delta x, \Delta y)|}{\sqrt{\Delta x^2 + \Delta y^2}} = 0$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{|g_2(\Delta x, \Delta y)|}{\sqrt{\Delta x^2 + \Delta y^2}} = 0 \Rightarrow \text{вектор } \begin{pmatrix} u \\ v \end{pmatrix} \text{ guarop. b m. } \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

нормир. вектор имеют вид

$$D_f(x_0, y_0) = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}, \quad \text{н.е. ван. век. к-Р}$$

$\Leftarrow \exists f\text{-}\mathbb{R}\text{-guarop. b m. } z_0 = x_0 + y_0 i$ и ван. век. к-Р.

$$\exists \frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0) = \alpha$$

$$\frac{\partial u}{\partial y}(x_0, y_0) = \frac{\partial v}{\partial x}(x_0, y_0) = \beta$$

$$\text{more } u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) = \alpha \Delta x - \beta \Delta y + \xi_1(\Delta x, \Delta y), \quad \text{wge}$$

$$\xi_1(\Delta x, \Delta y) = \bar{o}(\sqrt{\Delta x^2 + \Delta y^2}) \quad \begin{matrix} \Delta x \rightarrow 0 \\ \Delta y \rightarrow 0 \end{matrix}$$

$$v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0) = \beta \Delta x + \alpha \Delta y + \xi_2(\Delta x, \Delta y), \quad \text{wge}$$

$$\xi_2(\Delta x, \Delta y) = \bar{o}(\sqrt{\Delta x^2 + \Delta y^2}) \quad \begin{matrix} \Delta x \rightarrow 0 \\ \Delta y \rightarrow 0 \end{matrix}$$

$$\begin{aligned} f(z_0 + \Delta z) - f(z_0) &= (u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0)) + i(v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0)) = \\ &= \alpha \Delta x - \beta \Delta y + i(\beta \Delta x + \alpha \Delta y) + \underbrace{\xi_1(\Delta x, \Delta y) + i\xi_2(\Delta x, \Delta y)}_{\xi(\Delta z)} = (\alpha + \beta i)(\Delta x + i\Delta y) + \xi(\Delta z) = \end{aligned}$$

$$= A\Delta z + \xi(\Delta z), \quad \text{wge } A = \alpha + \beta i$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{|\xi_1(\Delta x, \Delta y)|}{\sqrt{\Delta x^2 + \Delta y^2}} = 0$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{|\xi_2(\Delta x, \Delta y)|}{\sqrt{\Delta x^2 + \Delta y^2}} = 0$$

$$|\xi(\Delta x, \Delta y)| \leq |\xi_1(\Delta x, \Delta y)| + |\xi_2(\Delta x, \Delta y)| \quad \text{неп-бо } \Delta$$

$$\Rightarrow \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{|\xi(\Delta x, \Delta y)|}{\sqrt{\Delta x^2 + \Delta y^2}} = 0$$

$$\Rightarrow f \text{ - } \mathbb{C}\text{-группа в м. } z_0 \quad \text{и} \quad f'(z_0) = A = \alpha + \beta i \quad \blacktriangle$$

$$f'(z_0) = \frac{\frac{\partial u(x_0, y_0)}{\partial x}}{\alpha} + \frac{\frac{\partial v(x_0, y_0)}{\partial x}}{\beta} i$$